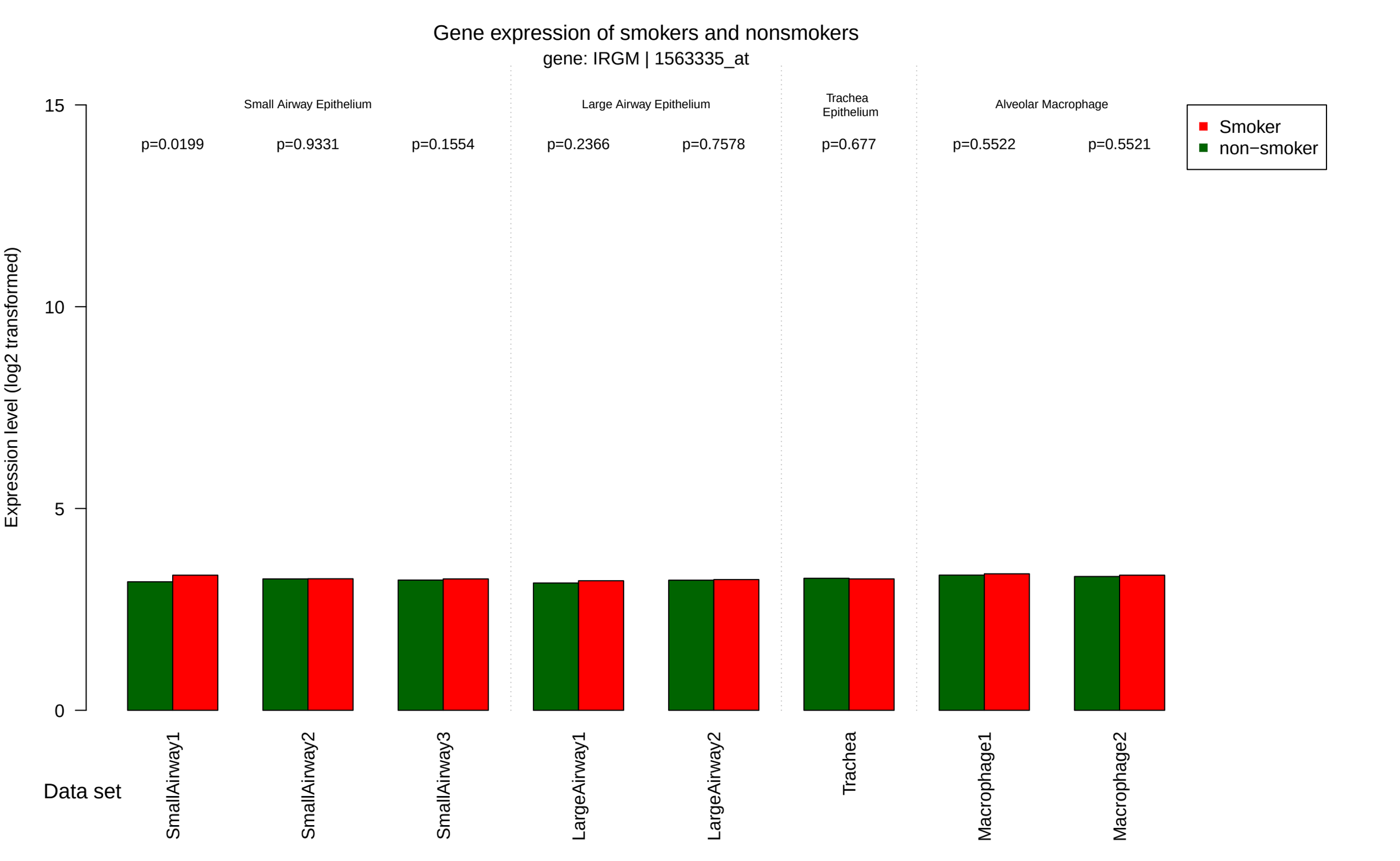
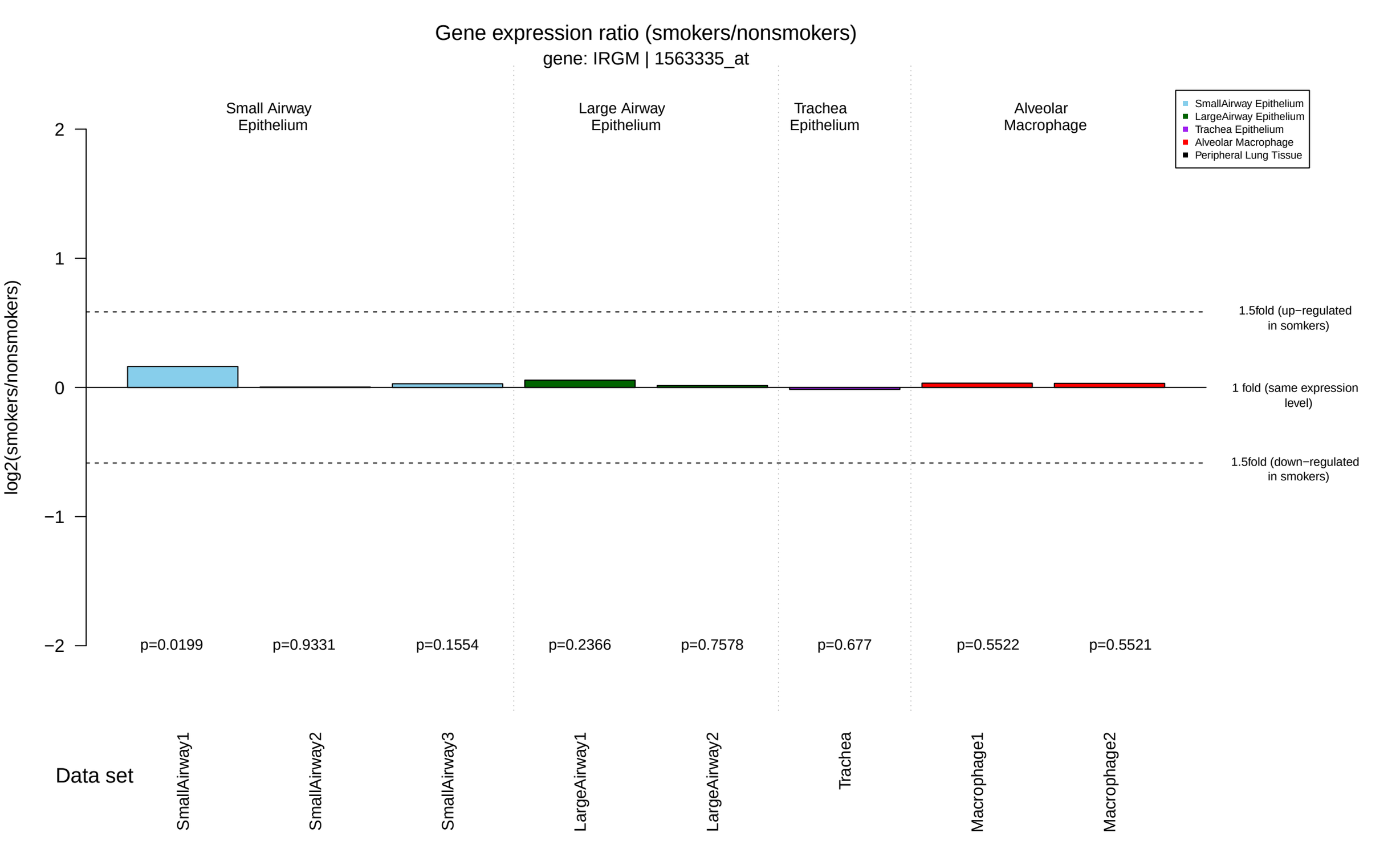
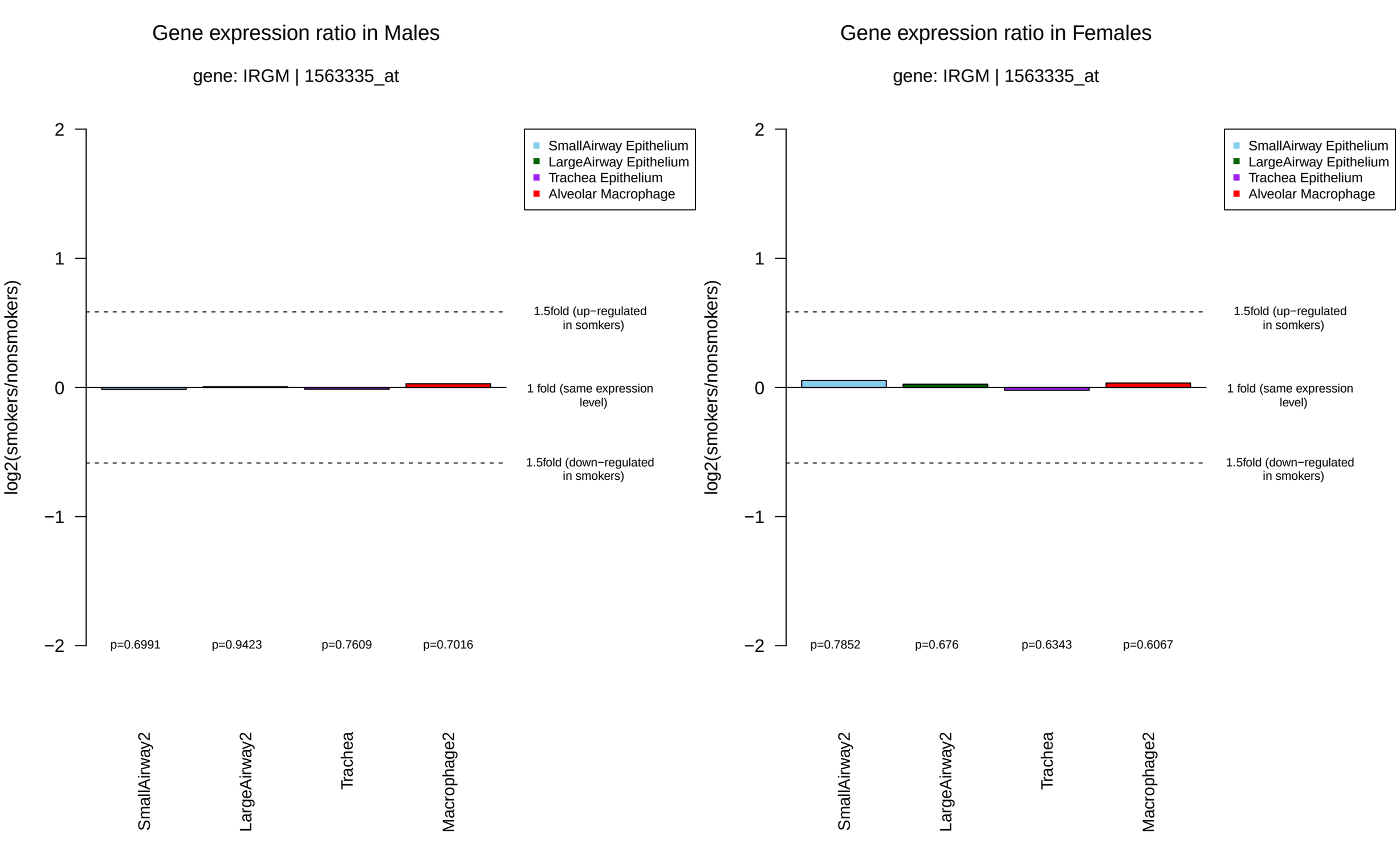
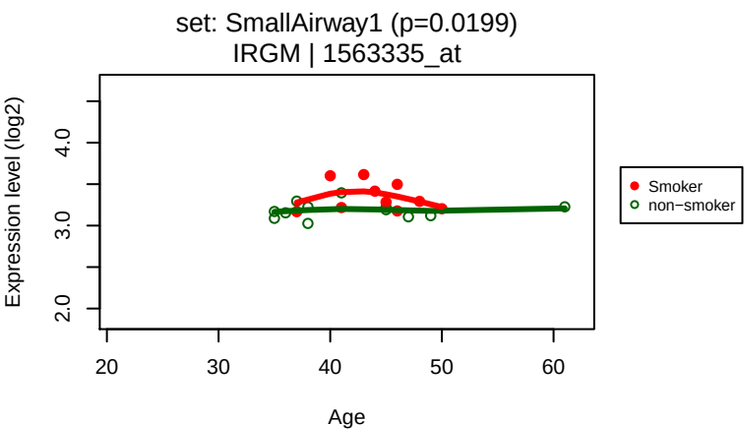
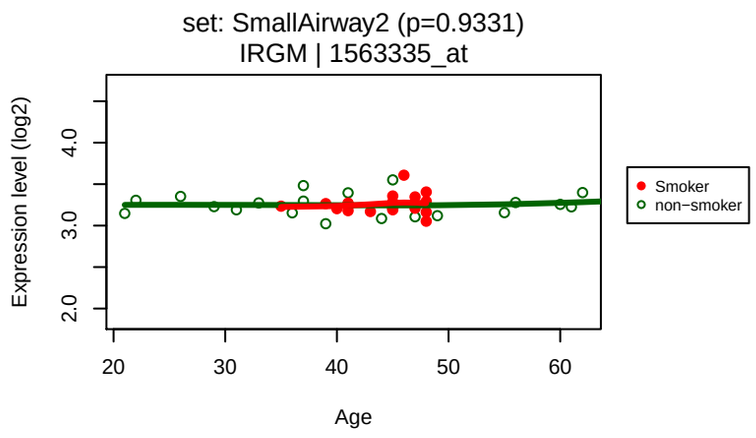
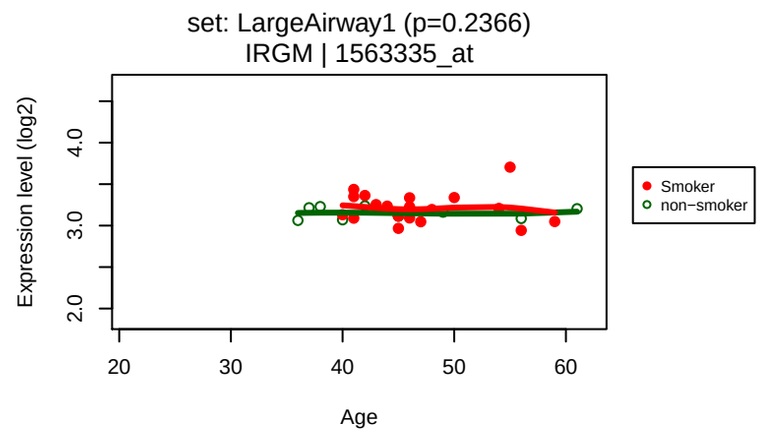

Gene expression of smokers and nonsmokers
gene: IRGM | 1563335_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0199
p=0.9331
p=0.1554
p=0.2366
p=0.7578
p=0.677
p=0.5522
p=0.5521
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

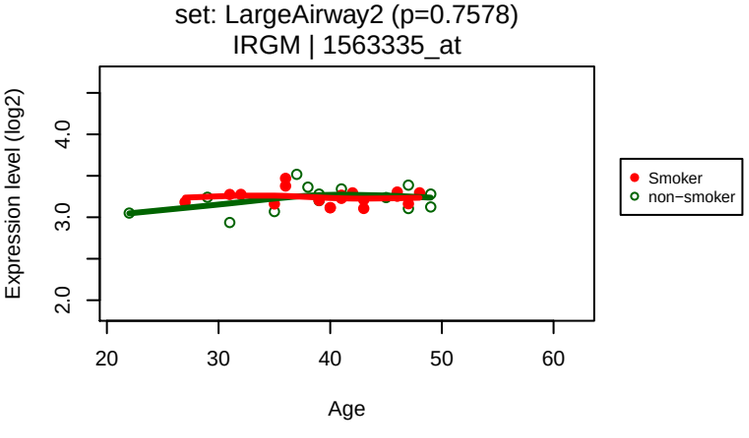

Gene expression ratio (smokers/nonsmokers)
gene: IRGM | 1563335_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0199
p=0.9331
p=0.1554
p=0.2366
p=0.7578
p=0.677
p=0.5522
p=0.5521
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

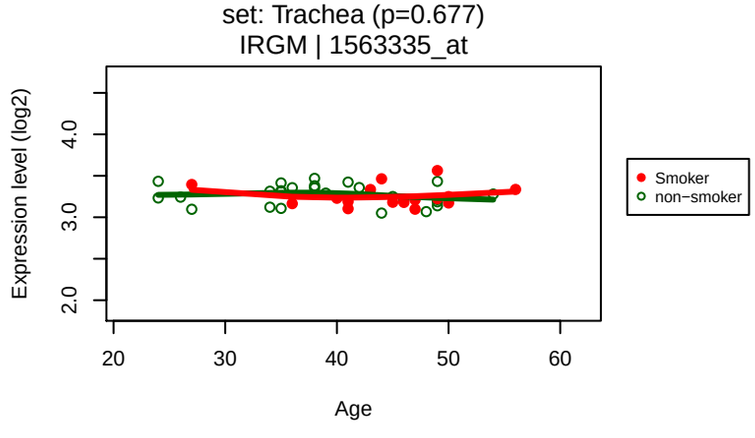

Gene expression ratio in Males
gene: IRGM | 1563335_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6991
p=0.9423
p=0.7609
p=0.7016
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRGM | 1563335_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7852
p=0.676
p=0.6343
p=0.6067
SmallAirway2
LargeAirway2
Trachea
Macrophage2

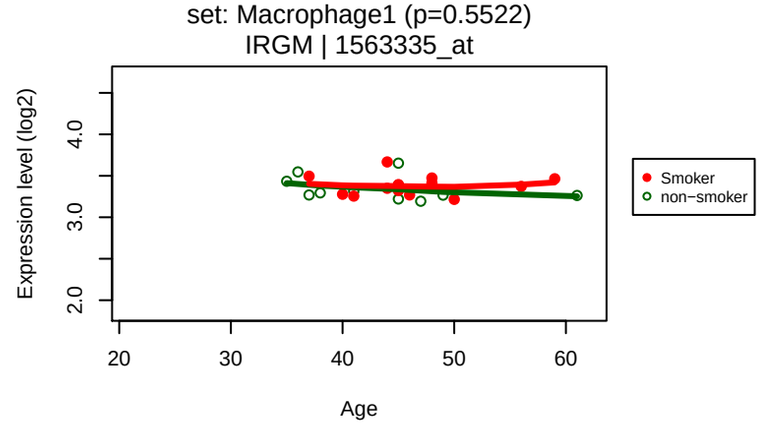

set: SmallAirway1 (p=0.0199)
IRGM | 1563335_at
Expression level (log2)
Age
Smoker
non−smoker

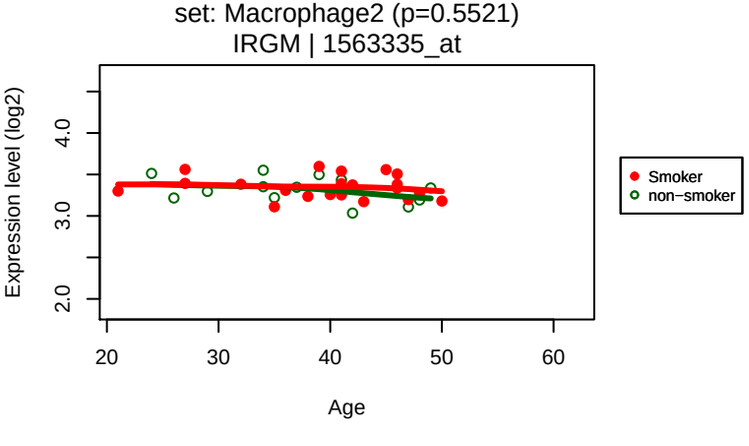

set: SmallAirway2 (p=0.9331)
IRGM | 1563335_at
Expression level (log2)
Age
Smoker
non−smoker

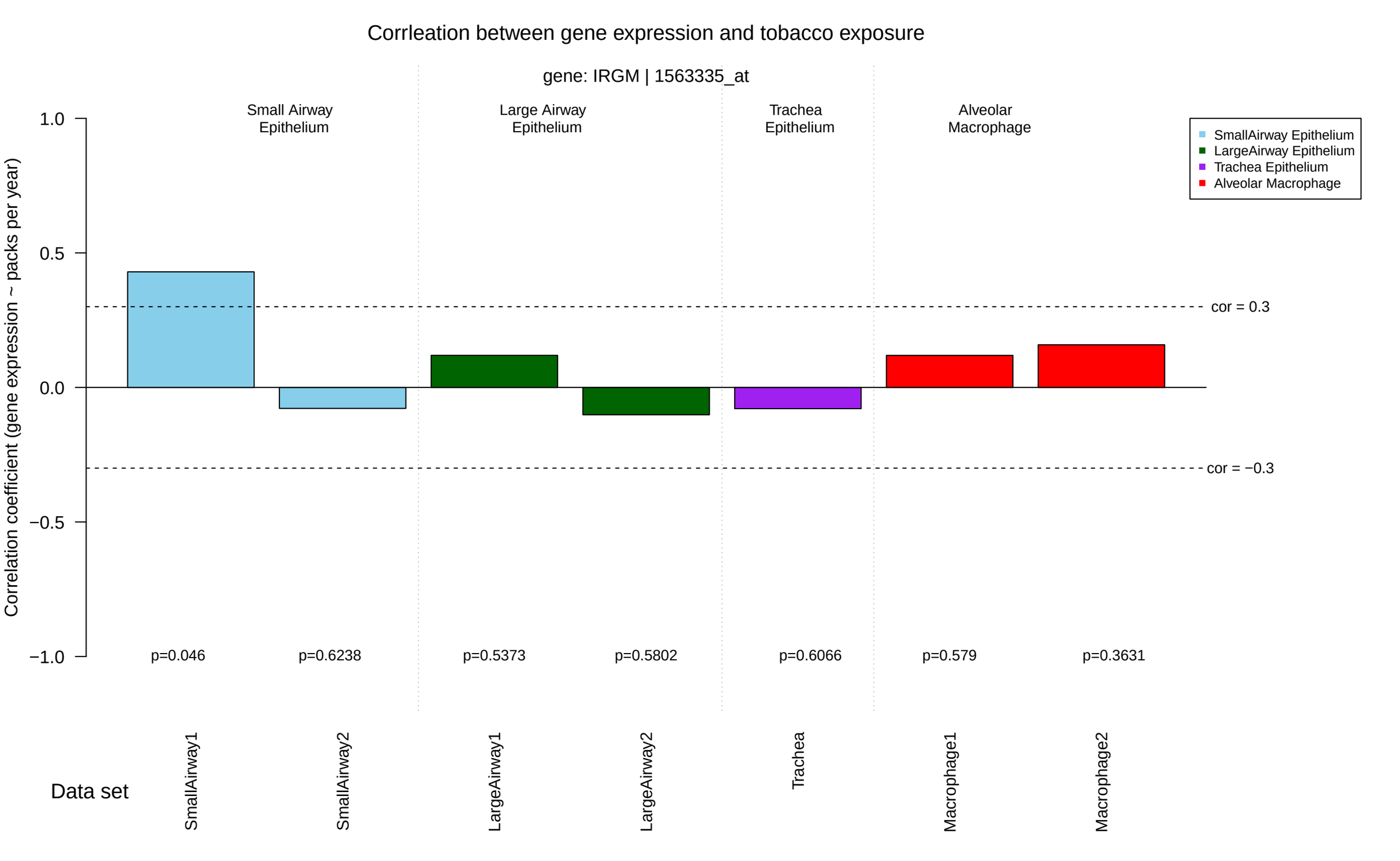

set: LargeAirway1 (p=0.2366)
IRGM | 1563335_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7578)
IRGM | 1563335_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.677)
IRGM | 1563335_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5522)
IRGM | 1563335_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5521)
IRGM | 1563335_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRGM | 1563335_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.046
p=0.6238
p=0.5373
p=0.5802
p=0.6066
p=0.579
p=0.3631
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2